Practical 3

Aim: Study Software Requirement Engineering. Student should include SRS document for current semester.

SRS(Software requirement specification):

- The production of the requirements stage of the software development process is Software Requirements Specifications (SRS) (also called a requirements document).
- This report lays a foundation for software engineering activities and is constructing when entire requirements are elicited and analyzed. SRS is a formal report, which acts as a representation of software that enables the customers to review whether it (SRS) is according to their requirements.
- Also, it comprises user requirements for a system as well as detailed specifications of the system requirements.
- The SRS is a specification for a specific software product, program, or set of applications that perform particular functions in a specific environment. It serves several goals depending on who is writing it.
- First, the SRS could be written by the client of a system. Second, the SRS could be written by a developer of the system.
- The two methods create entirely various situations and establish different purposes for the document altogether. The first case, SRS, is used to define the needs and expectation of the users. The second case, SRS, is written for various purposes and serves as a contract document between customer and developer.

SRS should address:

The basic issues that the SRS shall address are the following:

a) Functionality. What is the software supposed to do?
b) External interfaces. How does the software interact with people, the system's hardware, other hardware, and other software?

c) Performance. What is the speed, availability, response time, recovery time of various • software functions, etc.?

Name of System:

Online Trading System

Assumptions:

- Users must have valid bank accounts.
- Users must be registered before trading.
- Stock market API will provide real-time stock prices.
- Payment gateway will handle secure transactions.

Requirement 1: User Management

Req 1.1: Registration (Sign Up)

- Input: Name, Email, Phone Number, Bank Details, Password.
- Output: Confirmation message and User ID generated.
- Processing:

- System validates all user details.

- If details are correct, account is created.

- If error found, error message is displayed.

Req 1.2: Login

- Input: User ID and Password.
- Output: Access to trading dashboard.
- Processing:

  - Credentials are verified.

  - If valid, user is redirected to dashboard.

  - If invalid, error message displayed.

Req 1.3: Logout

- Input: Click logout button.
- Output: User session ends and redirected to homepage.

Requirement 2: Trading Management

Req 2.1: View Stock Market

- Input: Select stock or search stock name.
- Output: Real-time stock price and details displayed.
- Processing:

  - System fetches data from stock market API.

Req 2.2: Buy Stock

- Input: Select stock and quantity.
- Output: Confirmation message and transaction ID.
- Processing:

  - Check user balance.

  - If sufficient balance, transaction is completed.

  - If insufficient balance, error message displayed.

Req 2.3: Sell Stock

• Input: Select stock from portfolio and quantity.
• Output: Confirmation message and updated balance.
• Processing:

- Check stock availability in portfolio.
- If available, stock is sold.
- If not, error displayed.

Requirement 3: Portfolio Management

Req 3.1: View Portfolio

• Output: List of purchased stocks, quantity, current value, profit/loss.

Req 3.2: Transaction History

• Output: Detailed list of past transactions including buy/sell records.

Requirement 4: Payment System

Req 4.1: Add Funds

• Input: Amount and payment details (Card/UPI/Net Banking).
• Output: Balance updated and confirmation message.

Req 4.2: Withdraw Funds

• Input: Withdrawal amount.
• Output: Amount credited to bank account and confirmation message.

Requirement 5: Admin Management

Req 5.1: Manage Users

• Add or remove users.
• Monitor user activities.

Req 5.2: Manage Stocks

- Add stock listings.
- Remove stock listings.
- Monitor market data.

Requirement 6: Security

- User passwords stored in encrypted format.
- Secure payment gateway integration.
- SSL encryption for data transmission.
- Separate admin login panel.

Requirement 7: Help & Support

- User can contact support team.
- FAQ section available.

Requirement 8: E-Statements:

- Users can download transaction statements in PDF format.

Performance:

- System should respond within 2–3 seconds.
- Real-time stock updates without delay.

Reliability:

- System should be available 24/7.
- Automatic backup of data.

Security:

- Secure authentication mechanism.
- Protection against unauthorized access.

Portability:

System accessible on desktop, mobile, and tablet.
• Compatible with Windows, Linux, Android, iOS.

Usability:

• User-friendly interface.
• Easy navigation and simple design.

Conclusion:

• The SRS document clearly defines the requirements of the Online Trading System.
• It ensures clarity between customer and developer.
• It acts as a foundation for system design and implementatio

Practical 4

Aim: Study Software Requirement Engineering Student should include SRS document for current semester project.

Objectives:

1. Understand the requirements of the Online Trading System.
2. Define the scope, deliverables, and milestones of the project.
3. Identify resources, roles, and responsibilities.
4. Develop a project schedule and timeline.
5. Plan risk management, quality assurance, and communication strategies.
6. Create a comprehensive SPMP document.

Prerequisites:

1. Basic understanding of Software Engineering principles.
2. Knowledge of web development technologies (HTML, CSS, JavaScript, Backend).
3. Basic understanding of database management systems.
4. Familiarity with project management tools like Trello or Jira.

Tools and Resources:

1. Project Management Tools: Trello / Jira / Microsoft Project
2. Documentation Tools: MS Word / Google Docs
3. Development Tools:
   - Frontend: HTML, CSS, JavaScript
   - Backend: Node.js / Django / PHP
   - Database: MySQL / MongoDB
4. Version Control: Git & GitHub

Lab Procedure

Step 1: Project Overview

1. Problem Statement

Develop a secure and user-friendly **Online Trading System** that allows users to buy and sell stocks online, manage their portfolio, add/withdraw funds, and view real-time market data.

The system should include features like:

- User registration and authentication
- Real-time stock price display
- Buy/Sell functionality
- Portfolio management
- Payment gateway integration
- Admin monitoring system

2. Scope of the Project

Modules included:

1. User Management Module
2. Trading Module
3. Portfolio Module
4. Payment Module
5. Admin Module
6. Security Module

Boundary of Project:
The system will be developed as a **web application**.
Mobile version can be future enhancement.

3. Stakeholders

- Investors / Traders
- Admin
- Development Team
- Bank / Payment Gateway Provider
- Stock Market API Provider

Step 2: Project Deliverables and Milestones

1. Deliverables:

- SPMP Document
- SRS Document
- System Design Document
- Database Design
- UI Prototype
- Working Web Application
- Testing Report
- Final Documentation

2. Milestones:

Week	Activity
Week 1-2	Requirement Gathering & Analysis
Week 3-4	System Design & UI Prototype
Week 5-6	Frontend Development
Week 7-8	Backend & Database Development
Week 9	Integration & Testing
Week 10	Deployment & Documentation

Step 3: Resource Planning

1. Human Resources

- Project Manager – Oversees entire project
- Frontend Developer – Designs UI
- Backend Developer – Implements server logic
- Database Administrator – Manages database
- QA Engineer – Performs testing

2. Hardware and Software Resources

- Computer systems with internet access
- Web browser (Chrome/Firefox)
- Cloud hosting (AWS / Render / Heroku)
- Payment Gateway Sandbox

Step 4: Project Schedule

- Create Gantt chart using MS Project or Excel
- Define tasks for each module
- Allocate time for development and testing
- Identify dependencies (e.g., backend depends on database design)

Example Task Breakdown:

1. Requirement Analysis
2. System Design
3. Database Design
4. Frontend Coding
5. Backend Coding
6. API Integration
7. Testing
8. Deployment

Step 5: Risk Management

1. Identified Risks:

- Data security breach
- Server downtime
- API failure for stock data
- Delay in development
- Payment gateway errors

2. Mitigation Strategies:

- Use SSL encryption
- Regular data backups
- Use reliable stock API
- Maintain buffer time in schedule
- Test payment integration in sandbox mode

Step 6: Quality Assurance

1. Define coding standards.
2. Perform:
    - Unit Testing
    - Integration Testing
    - System Testing
    - User Acceptance Testing
3. QA Engineer will verify:
    - Accuracy of trading calculations
    - Security of transactions
    - Performance under load

Step 7: Communication Plan

1. Weekly team meetings
2. Communication via WhatsApp / Slack
3. GitHub for version control
4. Progress report submission every week

Step 8: Documentation

The SPMP document includes:

- Introduction (Purpose, Scope, Objectives)
- Project Organization (Team Structure, Roles)
- Managerial Process (Schedule, Risk Management, QA)
- Technical Process (Tools, Technologies, Methodology)
- Appendices (Glossary, References)

Expected Outcome:

1. A complete SPMP document for Online Trading System.
2. Clear project roadmap with defined milestones.
3. Risk management and quality assurance plan.

Practical 5

Aim: Do Cost and Effort Estimation using different Software Cost Estimation models.

Objectives:

- To understand the requirements of the Online Trading System.
- To apply COCOMO model for effort and time estimation.
- To apply Function Point Analysis for size-based estimation.
- To use Expert Judgment for practical cost prediction.
- To compare results from different estimation techniques.

Prerequisites:

- Basic knowledge of Software Engineering principles.
- Understanding of COCOMO and Function Point Analysis models.
- Knowledge of web-based application development.
- Familiarity with spreadsheet tools like Excel.

Requirements of Online Trading System

1. User Management Module

- User registration
- User login and logout
- Authentication and authorization
- Profile management

2. Trading Module

- Real-time stock price display
- Buy stock functionality
- Sell stock functionality
- Order processing system

3. Portfolio Module

- View stock holdings
- Profit and loss calculation
- Transaction history

4. Payment Module

- Add funds
- Withdraw funds
- Payment gateway integration

5. Admin Module

- Monitor users
- Manage stock listings
- Generate system reports

6. Security Module

- SSL encryption
- Secure transactions
- Data protection mechanisms

Software Cost Estimation Models

1. COCOMO Model:

Project Classification

- Project Type: Semi-Detached
- Reason: Moderate complexity and experienced development team

Size Estimation

- Estimated Size: 25 KLOC

Effort Calculation

- Formula: $E = a \times (KLOC)^b$
- For Semi-Detached: $a = 3.0$, $b = 1.12$
- $E = 3.0 \times (25)^{1.12}$
- Effort ≈ 96 Person-Months

Development Time Calculation

- Formula: $D = 2.5 \times (E)^{0.35}$
- Development Time ≈ 15 Months

Cost Estimation

- Cost per Person-Month = ₹80,000
- Total Cost = 96 × 80,000
- Total Cost ≈ ₹76,80,000

2. Function Point Analysis (FPA):

Functional Components Identified

- External Inputs (Login, Buy, Sell, Add Funds)
- External Outputs (Portfolio Reports)
- External Inquiries (Stock Search)
- Internal Logical Files (User Data, Transaction Records)
- External Interfaces (Stock API, Payment Gateway)

Size Calculation

- Unadjusted Function Points (UFP) = 220
- Value Adjustment Factor (VAF) = 1.15
- Adjusted Function Points = 220 × 1.15
- Adjusted FP = 253

Effort Estimation

- Effort per Function Point = 18 hours
- Total Effort = 253 × 18
- Total Effort = 4554 hours

Convert to Person-Months

- 1 Person-Month = 160 hours
- Person-Months = 4554 ÷ 160
- Person-Months ≈ 28.5

Cost Estimation

- Cost per hour = ₹500
- Total Cost = 4554 × 500
- Total Cost ≈ ₹22,77,000

3. Expert Judgment:

Expert Estimates

- Expert 1: ₹25,00,000 – 12 months
- Expert 2: ₹30,00,000 – 14 months
- Expert 3: ₹28,00,000 – 13 months

Average Estimation

- Average Cost ≈ ₹27,66,000
- Average Development Time ≈ 13 Months

Comparison and Analysis:

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- COCOMO provides higher estimation due to complexity consideration.
- FPA provides estimation based on system functionality.
- Expert Judgment provides practical industry-based estimation.
- FPA and Expert Judgment give more realistic results for this project.

Expected Result:

- Detailed cost and effort estimation report.
- Application of three estimation techniques.
- Comparison of effort, time, and cost.

Conclusion

- Cost and effort estimation is essential for project planning.
- Different models provide different results based on approach.
- Online Trading System estimation shows variation across models.
- Combination of FPA and Expert Judgment is most suitable for this project.